



WEEKLY INTERNSHIP PROGRESS REPORT

Module: Week 3 – Intermediate Laravel (Student Portal)

Submitted By: Muneeb ur Rehman

Role: Full-Stack Web Developer Intern

Stack/Focus: PHP, Laravel, MySQL, Bootstrap 5

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1. Executive Summary

The primary objective of Week 3 was to advance into intermediate Laravel architecture by developing a secure, relational **Student Portal**. The week focused on implementing robust authentication, role-based authorization, relational databases (Eloquent ORM), and secure file handling. The module culminated in a fully integrated administrative dashboard featuring a global UI layout, seamless routing, and complete CRUD operations with image upload capabilities.

2. Key Achievements & Task Breakdown

- **Day 11: Eloquent Relationships:** Transitioned the database from a flat structure to a relational architecture. Implemented a `One-to-Many` Eloquent relationship mapping the `Course` model to the `Student` model, utilizing foreign keys (`course_id`) to ensure data normalization.
- **Day 12: Authentication:** Scaffolding complete user registration, login, and session management systems using **Laravel Breeze**.
- **Day 13: Middleware & Authorization:** Engineered custom role-based access control (RBAC). Created an `AdminMiddleware` layer to intercept incoming HTTP requests, verifying active sessions and user roles to protect backend administrative routes from unauthorized access.
- **Day 14: File Uploads & Storage:** Implemented `multipart/form-data` processing within the `StudentController`. Configured Laravel's local storage disk and symbolic linking (`storage:link`) to securely validate, store, and publicly serve student profile pictures.
- **Day 15: Module Integration & UI Architecture:** Unified the backend logic with a global frontend architecture. Implemented Eloquent Eager Loading (`with('course')`) to optimize database queries, built a master Blade layout with a global navigation bar, and established automatic route redirection for a seamless user experience.

3. Technical Implementation Highlights

- **Eager Loading:** Replaced `N+1` query vulnerabilities with Eloquent's `with()` method, ensuring related course data is fetched simultaneously with the paginated student dataset to optimize server performance.
- **Master Layouts:** Transitioned from isolated HTML files to a centralized `@extends('layouts.master')` Blade architecture. This allowed for a single source of truth for the global navigation bar and Bootstrap 5 CDN integration, drastically reducing code duplication.
- **Secure File Validation:** Enforced strict server-side validation rules (`image|mimes:jpeg,png,jpg|max:2048`) before processing file streams, ensuring server integrity against malicious script uploads.

4. Challenges Encountered & Resolutions

- **Environment Configuration Caching:** Encountered a database connection mismatch where Laravel defaulted to SQLite despite `.env` variables being mapped to MySQL.
 - *Resolution:* Executed `php artisan config:clear` to purge cached configurations, followed by `php artisan migrate:fresh` to successfully reconstruct the database schema within the XAMPP MySQL environment.

- **Routing & UI Flow Disconnection:** The initial architectural setup lacked front-facing entry points, requiring manual URL entry to access administrative modules.
 - *Resolution:* Re-engineered `routes/web.php` to include root-level redirects and overriding Laravel Breeze's default `/dashboard` logic. Integrated a global navigation UI to ensure all modules were accessible via standard GUI interactions.

5. Conclusion

Week 3 successfully elevated the application from a basic CRUD script to a production-ready, secure portal. Mastering Eloquent relationships, middleware protection, and file storage has solidified a strong understanding of Laravel's intermediate capabilities, establishing a highly functional and secure baseline for future development sprints

Student Portal

HomeDashboardStudents ListAdd Student

Logout

Student Management

A clearer way to manage your students.

Keep student records, courses, and profile details organized in one simple admin portal.

Login to Admin Portal

Admin Portal

Welcome to the admin dashboard.

Manage student records and course assignments from one place.

View Students

Add Student

Students List

Browse all registered students and their courses.

Add Student

Avatar	Name	Email	Course
	Muneeb Ur Rehman	awanzada1289@gmail.com	Laravel Advanced
	safety gaurd 70 (n)	urrehmanmehboob709@gmail.com	Laravel Advanced

Add New Student

Create a student record and assign a course.

Full Name

Email Address

Phone

Assign Course

-- Select a Course --



Profile Picture

Choose file No file chosen

Add New Student

Create a student record and assign a course.

Student and Profile Picture Saved Successfully!